



Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake using Spray Patternator

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What happens when the precision of agricultural spraying collides with the turbulence of a propeller wake? This study answers that question by investigating the impact of propeller wake on the spray geometry of standard ASABE agricultural nozzles. A 3D spray patternator was designed, built, and integrated into the University of Dayton Spray Test Section to capture spray patterns both with and without the influence of a propeller. The nozzles were positioned directly beneath the hub of the propeller, and spray patterns were captured from the nozzle tip to the bottom of the spray at 1.4-inch intervals. These images were used to reconstruct and quantify the spray geometry, including spray width, depth, and droplet distribution. The results reveal that finer nozzles, such as the 11001 and 11003, experienced up to a 100% increase in spray width and a 50% reduction in depth, with significant deformation of the flat-fan geometry. Coarser nozzles, like the 8008, showed minimal changes. Quantification of droplet dynamics indicated that finer nozzles, with higher drag coefficients, are more vulnerable to propeller-induced turbulence.

Nomenclature

W	Weight of the drone	V_0	Rotor inflow velocity
n	Number of propellers	D	Discharge Coefficient
ρ	Fluid density	Re	Reynolds Number
A	Rotor disk area	$D_{Orifice}$	Nozzle Orifice Diameter
C_d	Drag Coefficient	$A_{Orifice}$	Nozzle Orifice Area
$V_{Nozzle Exit}$	Droplet Exit Velocity from Nozzle	D_{V_X}	Volume diameter with X% of liquid atomized into droplets below this value
$V_{Propeller Exit}$	Exit Velocity from Propeller	R_{prop}	Radius of the propeller(s)
Q	Volumetric Flow Rate		
GPM	Gallons Per Minute		

I. Introduction

As of 2020, nearly 110,000 UAVs were being used for agricultural spraying, covering an operational area of over 60 million hectares in China alone [1]. In Japan, more than 50% of rice paddies are treated using UAVs [2]. For spray applications involving multi-rotor drones, it is crucial to account for the influence of the drone's wake on droplet size distribution from the nozzles and the potential for spray drift. Spray drift is a major issue in agricultural chemical applications, potentially causing damage to neighboring crops or contamination of nearby water sources, schools, or

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hospitals [3]. This issue is influenced by several factors, including the nozzle type and configuration, drone altitude, forward speed, and the chemical composition being sprayed. In agricultural drones, nozzles are typically positioned beneath the drone body, as shown in Figure 1. Some drones have nozzles mounted directly below the propellers (Figure 1a), while others feature booms with nozzles spaced evenly for spraying (Figure 1b). Certain drones are equipped with extended booms to reduce wake interaction, as illustrated in Figure 1c. Since the nozzles are positioned beneath the propellers, the interaction between wake velocity and droplets cannot be ignored. The wake of a propeller is highly unsteady and three-dimensional in nature, although its complexity can be simplified by analyzing the mean and fluctuating components separately. Due to their smaller size, droplet trajectories and velocities are likely to align closely with the propeller wake's flow patterns.

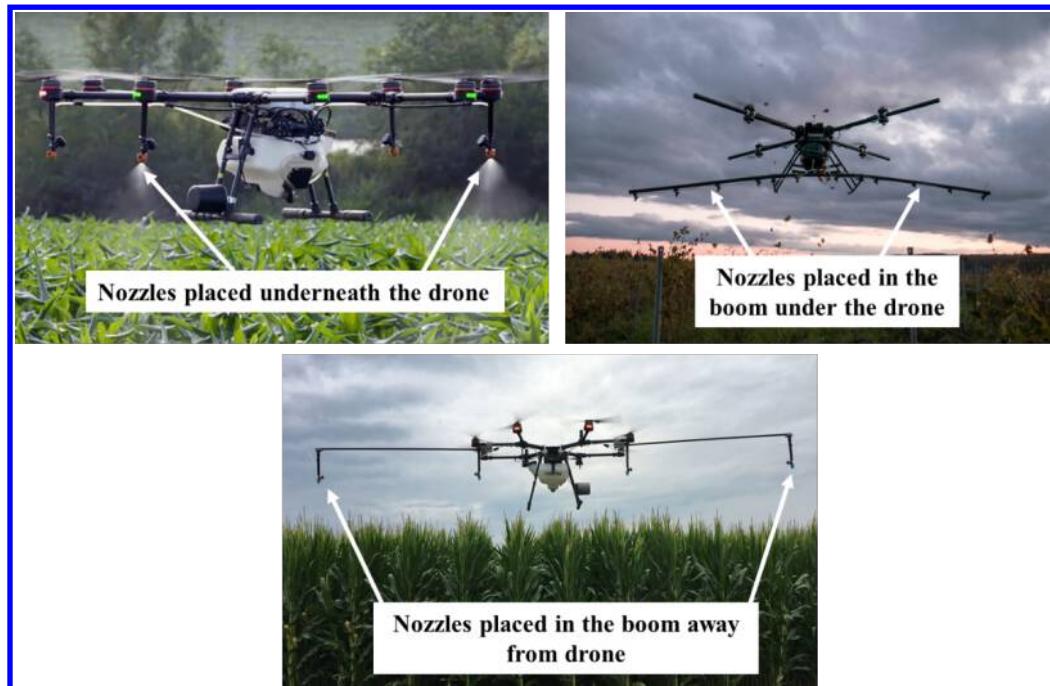


Figure 1 Agri-drones where nozzles are placed a) underneath the propellers [4] b) in the boom under the drone [5] c) in the boom away from the propellers [6]

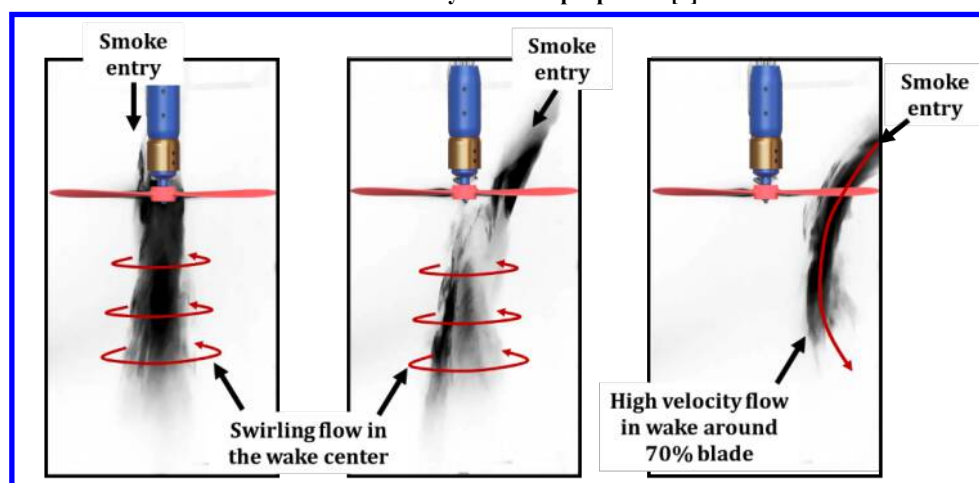


Figure 2 Smoke flow visualization of propeller wake illustrating (a) swirling flow near the center of the wake, (b) reduced swirling near blade tips, and (c) high-velocity flow regions around 70% blade radius.

A smoke flow visualization of the propeller wake is shown in Figure 2. Smoke is injected upstream of the propeller at various blade radius locations, and the resulting trajectories of the smoke lines are visualized. The flow characteristics and trajectories observed in Figure 2 are representative of all propellers used in agricultural drones. One critical characteristic of the propeller wake is its highly three-dimensional nature, particularly beneath the propeller hub where nozzles are often positioned, as illustrated in Figure 1. This three-dimensional swirling flow can significantly influence the atomization and droplet size distribution of the spray due to increased rotation and shear forces. The swirl component is more pronounced near the hub and diminishes at the blade tips.

Cheng et al. [7] used hot-wire anemometry to measure velocity gradients in the propeller wake, reporting significant droplet breakup due to steep velocity gradients and intense downwash. Yang et al. [8] employed computational simulations to study droplet trajectories from a centrifugal nozzle in the wake of a 6-rotor UAV, revealing substantial drift and spiral descent of droplets, with wind speed exerting a greater influence on trajectories than propeller downwash at specific heights. Similarly, Wang et al. [9] assessed droplet size and drift potential in a wind tunnel with various nozzle types and RPMs, showing that specific nozzle designs can reduce drift significantly while higher wind speeds increase drift. Grant et al. [10] corroborated these findings, identifying wind speed as the dominant factor influencing droplet drift, with propeller RPM having a secondary effect. Further, Wang et al. [11] demonstrated through wind tunnel tests that wind speed and flying height greatly affect deposition and drift areas, while Berner et al. [12, 13] observed enhanced penetration into crop canopies caused by propeller wake, with effects varying based on plant height. Tang et al. [14] utilized particle image velocimetry to study spray swath alterations, concluding that propeller wake-induced low-pressure zones can broaden deposition areas by up to 150%.

Tierney et al. [15] conducted a series of investigations, including laser diffraction, gutter volume collection, and high-speed shadowgraphy, to quantify the effects of propeller wake on agricultural sprays. Their findings indicated that droplet size measurements obtained through line-of-sight laser diffraction are significantly biased. The influence of the propeller wake skews the results toward a droplet population with a higher mean droplet diameter. Shadowgraph and flow visualization experiments revealed pronounced oscillations in the spray, along with substantial twisting and deformation of the flat-fan spray geometry, as shown in Figure 3.

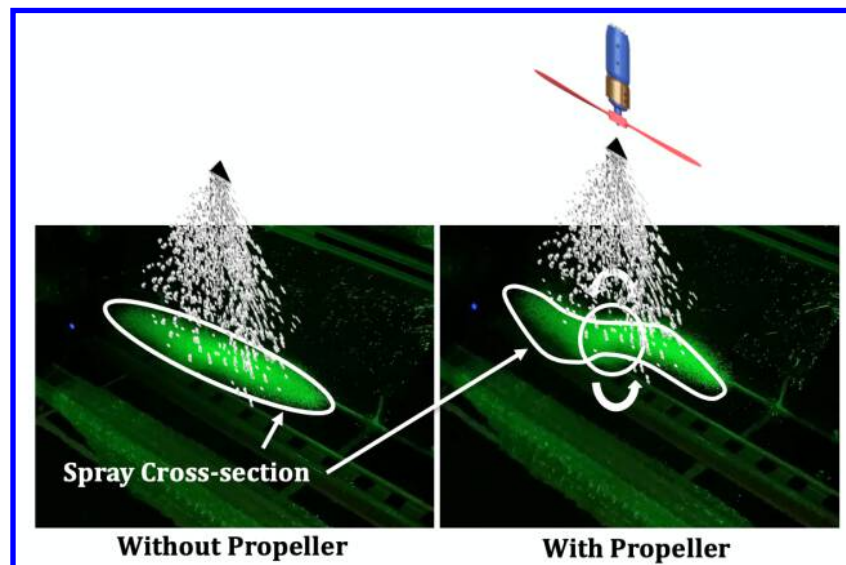


Figure 3 Comparison of spray geometry without propeller influence (left) and with propeller influence (right). The presence of the propeller introduces significant twisting, oscillation, and deformation of the spray cross-section

The highly unsteady and three-dimensional interactions between the propeller wake and spray create significant challenges in accurately characterizing their effects on nozzle spray geometry. To address this, the current study involves the design, construction, and integration of a spray patternator into the University of Dayton Spray Test

rig. This new approach aims to comprehensively characterize the three-dimensional influence of the propeller wake on standard agricultural nozzle sprays.

II. Experimental setup

A. Spray Test Stand

All experiments were conducted in the University of Dayton Wind Tunnel Laboratory (UD-LSWT) using the propeller-spray test stand depicted in Figure 4. The test stand accommodates one or two APC 17x7 propellers mounted on parallel linear traverses, with a traversing distance of 1 meter. Pipe fittings were installed to position standard agricultural nozzles within the propeller wake at varying transverse and downstream distances. Standard ASABE nozzles, classified according to the Fritz ASABE nozzle classifications [16], were utilized for this study. The nozzles were supplied with constant pressure using a Burcam constant pressure pump, and pressure measurements were obtained with an SSI Technologies model MG1-200-A-9V-R digital pressure gauge, accurate to $\pm 0.25\%$. The nozzles were categorized based on spray angle, volumetric flow rate, and droplet size, ranging from very fine to extremely coarse. For instance, the "11001" nozzle has a spray angle of 110 degrees and an expected output of 0.1 gallons per minute. The specifications for all tested nozzles are detailed in Table 1. At the rated pressure of 21.7 psi, the reported volumetric flow rate for the 6515 nozzle was measured as 1.15 GPM, which differs from the 1.5 GPM specified in the Fritz ASABE classifications [16]. All tests were conducted using water as the spraying medium. The laser diffraction, gutter tests, and high-speed shadowgraph experiments were performed using the test stand, and the corresponding methodologies are detailed in the following subsections.

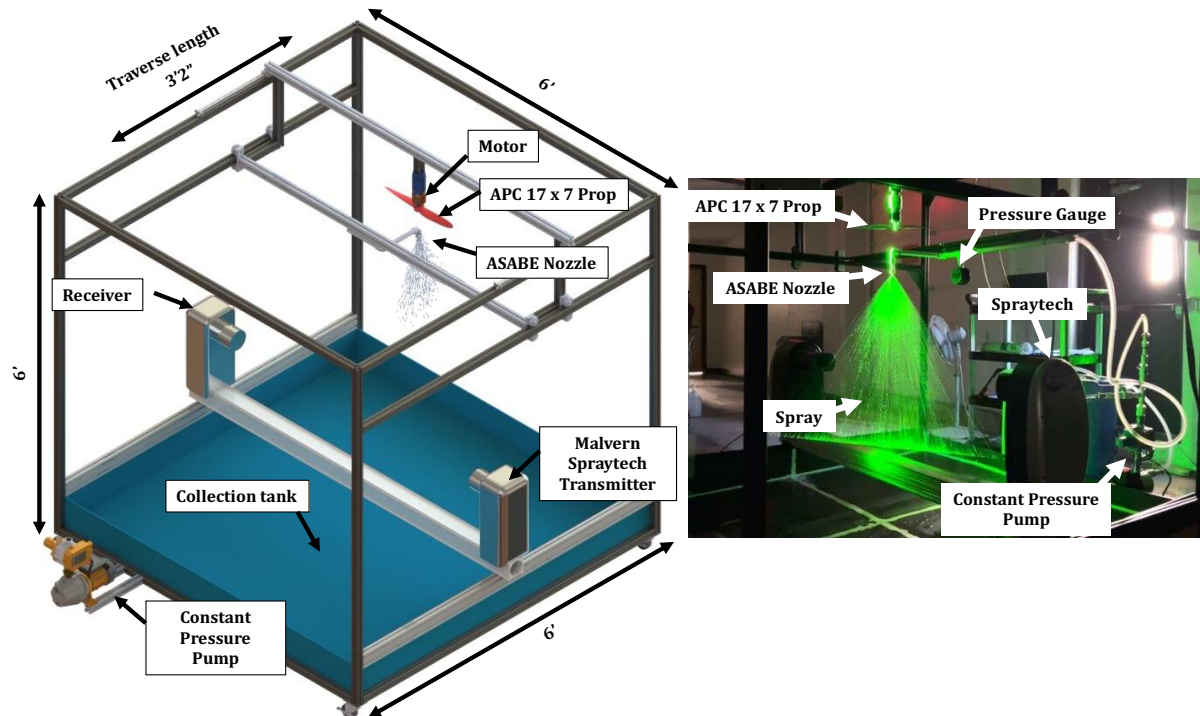


Figure 4 Propeller-spray test stand used for experiments. (Left) Schematic showing the setup with key components, including APC 17x7 propeller, ASABE nozzle, and Malvern Spraytech laser diffraction system. (Right) Photograph of the operational setup, highlighting spray visualization.

B. Propeller

All nozzles were tested at a baseline of 0 propeller RPM and at 4900 RPM. Thrust, propeller RPM, and throttle settings were determined using the propeller performance tests conducted at UD-LSWT, as reported by Cai et al. [17]. The exit velocity from the propeller was estimated using Equation 1, derived from actuator disk theory [18]:

$$V_{Propeller Exit} = \sqrt{\frac{T}{2\rho_{\infty}A}} \quad (1)$$

Where $V_{Propeller Exit}$ represents the exit velocity by the propeller, T represents the measured thrust, and A represents the rotor disk area calculated from the propeller diameter. Table 2 summarizes the chosen throttle settings, the corresponding thrust, and the estimated exit velocities. The thrust values at 3900, 4900, and 5900 RPM for the APC 17x7 propeller produced exit velocities similar to those of DJI UAVs, which generate induced velocities ranging from 5 to 15 m/s.

Table 1 Specifications of the ASABE Standard Nozzles [16]

Nozzle	Fan Angle (Deg)	Volumetric Flow Rate (GPM)	Pressure rating (PSI)	Dv(10) (μm)	Dv(50) (μm)	Dv(90) (μm)
11001 (very fine)	110	0.1	65.3	61.9	133.9	229.3
11003 (fine)	110	0.3	43.5	108.8	242.8	405.7
11006 (medium)	110	0.6	29	158.8	348.3	571.1
8008 (coarse)	80	0.8	36.3	197	433.6	630.5
6510 (very coarse)	65	1.0	29	235.4	516.7	850.1
6515 (extremely coarse)	65	1.15	21.7	315.4	673.5	1116.6

Table 2 Thrust, throttle settings, propeller RPM, and estimated exit velocities of the APC 17x7 propeller

Thrust (N)	Throttle Setting (%)	RPM	Density (kg/m ³)	$V_{Propeller Exit}$ (m/s)
20.53	40	3900	1.18	7.5
23.62	50	4900		8.3
35.63	65	5900		10.1

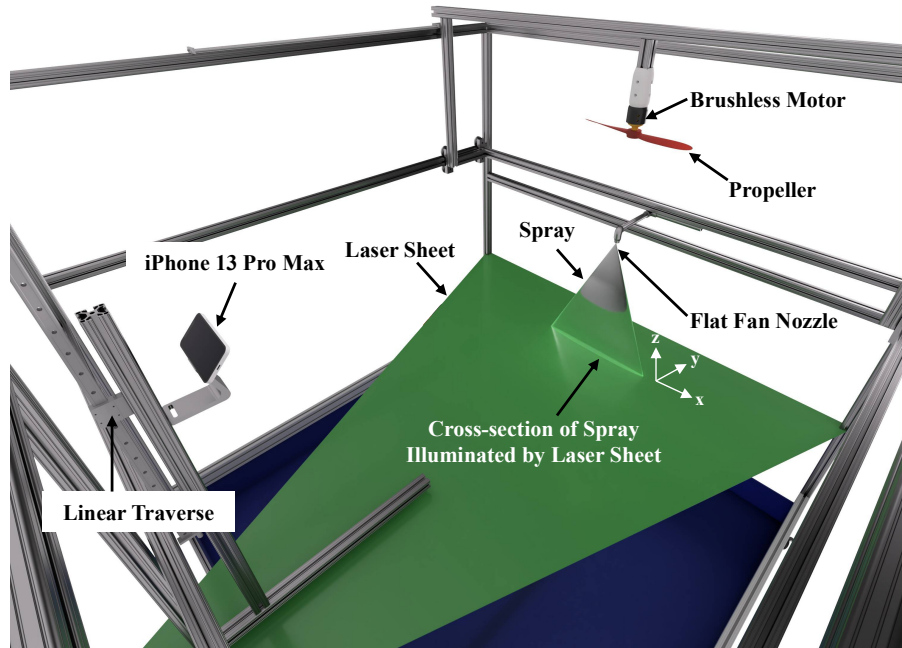


Figure 5 Spray patternator setup showing the alignment of the 532 nm laser, flat fan nozzle, and iPhone 13 Pro Max camera to capture the cross-sectional geometry of the illuminated spray

C. Spray Patternator

The spray patternator utilized a 1W 532 nm laser from OX Lasers, aligned perpendicular to the plane of the flat fan spray, to illuminate the cross-sectional geometry of the spray from the nozzle, as shown in Figure 5. Multiple iterations of lasers with varying power levels and configurations were tested to achieve adequate intensity and coverage for capturing the spray volume. The chosen setup used a 1W laser in combination with two plano-concave lenses with focal lengths of -20 mm and -7 mm, which provided sufficient intensity and wide coverage for the spray, particularly for the 110-degree nozzles (11001, 11003, and 11006). Several iterations with different cameras were performed to capture the illuminated spray with the required field of view, depth of field, and aperture. Ultimately, an iPhone 13 Pro Max, equipped with a 12 MP sensor, $1.9\ \mu\text{m}$ pixels, and a 26 mm equivalent $f/1.5$ aperture lens, was selected to capture the backscatter from the illuminated spray droplets. Both the laser and the camera were mounted on a single vertical traverse, which allowed the cross-section of the spray to be captured at different heights from the nozzle. A sensitivity study was conducted to determine the optimal angle for the camera to capture the maximum intensity of backscattered light. The test matrix for capturing the geometry of the spray using the patternator is presented in Table 3, and the schematic of the planes captured is illustrated in Figure 6. The patternator was employed to record a 5-second video of the spray cross-section at 30 fps, starting at a distance of 10 inches from the nozzle and progressing down to 38 inches in 1.4-inch increments, as depicted in Figure 6. Each nozzle was tested under two conditions: without the propeller and with the propeller operating at 4900 RPM.

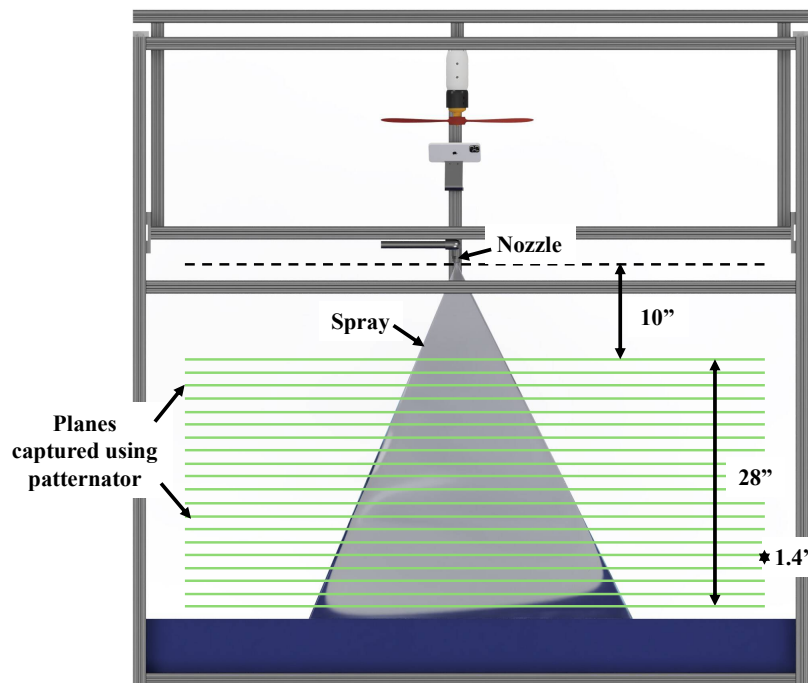


Figure 6 Schematic showing the planes captured using the spray patternator, starting 10 inches from the nozzle and progressing in 1.4-inch increments

D. Workflow for Processing Data from Patternator

The workflow for generating a 3D reconstruction of the nozzle spray geometry is shown in Figure 7. Due to the angular placement of the camera relative to the laser light sheet, perspective correction was necessary to account for parallax errors. The images before and after applying perspective correction are displayed in Figure 7. After calibrating the field of view, the laser sheet and camera were traversed at different vertical distances along the nozzle spray. Images were captured at 30 fps for 5 seconds at each vertical location and then averaged to produce a single image for each

vertical position. Each averaged image underwent perspective correction and was interpolated to reconstruct the 3D spray geometry

Table 3 Test matrix used during patternator experiments

Nozzle	Propeller RPM	Duration of Data Acquisition (s)	Frame Rate	Field of View (m)	Distance from Nozzle (m)
11001 (very fine)	0, 4900	5	30	1.3 x 2	0.25 to 0.96 in 0.03 increments
11003 (fine)					
11006 (medium)					
8008 (coarse)					
6510 (very coarse)					
6515 (extremely coarse)					

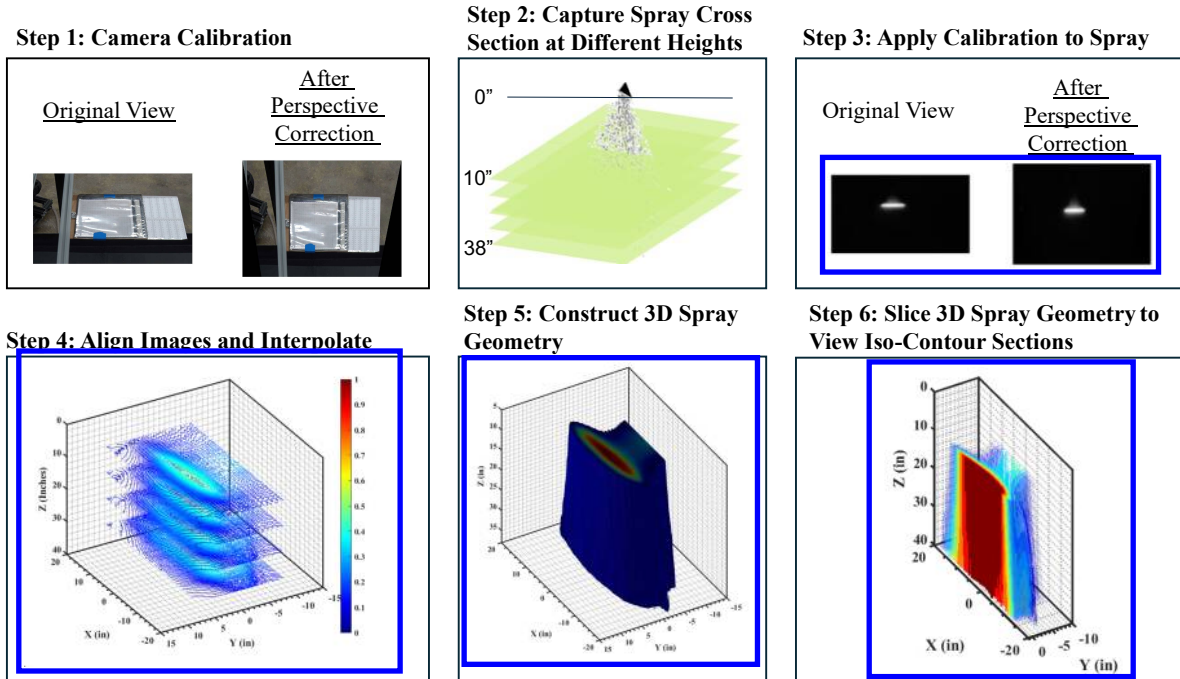


Figure 7 Workflow for processing patternator data, including raw and perspective-corrected images (top row), planes captured at different vertical positions (middle column in first row), and 3D reconstruction of nozzle spray geometry (bottom row)

III. Results

The results from the patternator experiments for all tested nozzles, with and without the influence of the propeller, are shown in Figure 8. The first two rows correspond to the coarser nozzles (6515, 6510, and 8008), while the last two rows represent the finer nozzles (11006, 11003, and 11001). The first two digits of the four-digit nozzle codes and the first three digits of the five-digit nozzle codes indicate the spray angle of the flat fan spray. For instance, the 6510 nozzle has a spray angle of 65°, whereas the 11006 nozzle has a spray angle of 110°. The 3D reconstructions of the spray geometries were visualized using iso-surfaces of intensity values ranging from 0.15 to 0.45 for coarser nozzles and 0.05 to 0.35 for finer nozzles. These intensity ranges account for the lower droplet densities and smaller droplet sizes produced by finer nozzles compared to coarser ones. Transparency was applied to the iso-surfaces, with higher transparency near the edges of the spray and lower transparency near the high-intensity regions at the center. This transparency allows for better visualization of the higher droplet density regions within the spray.

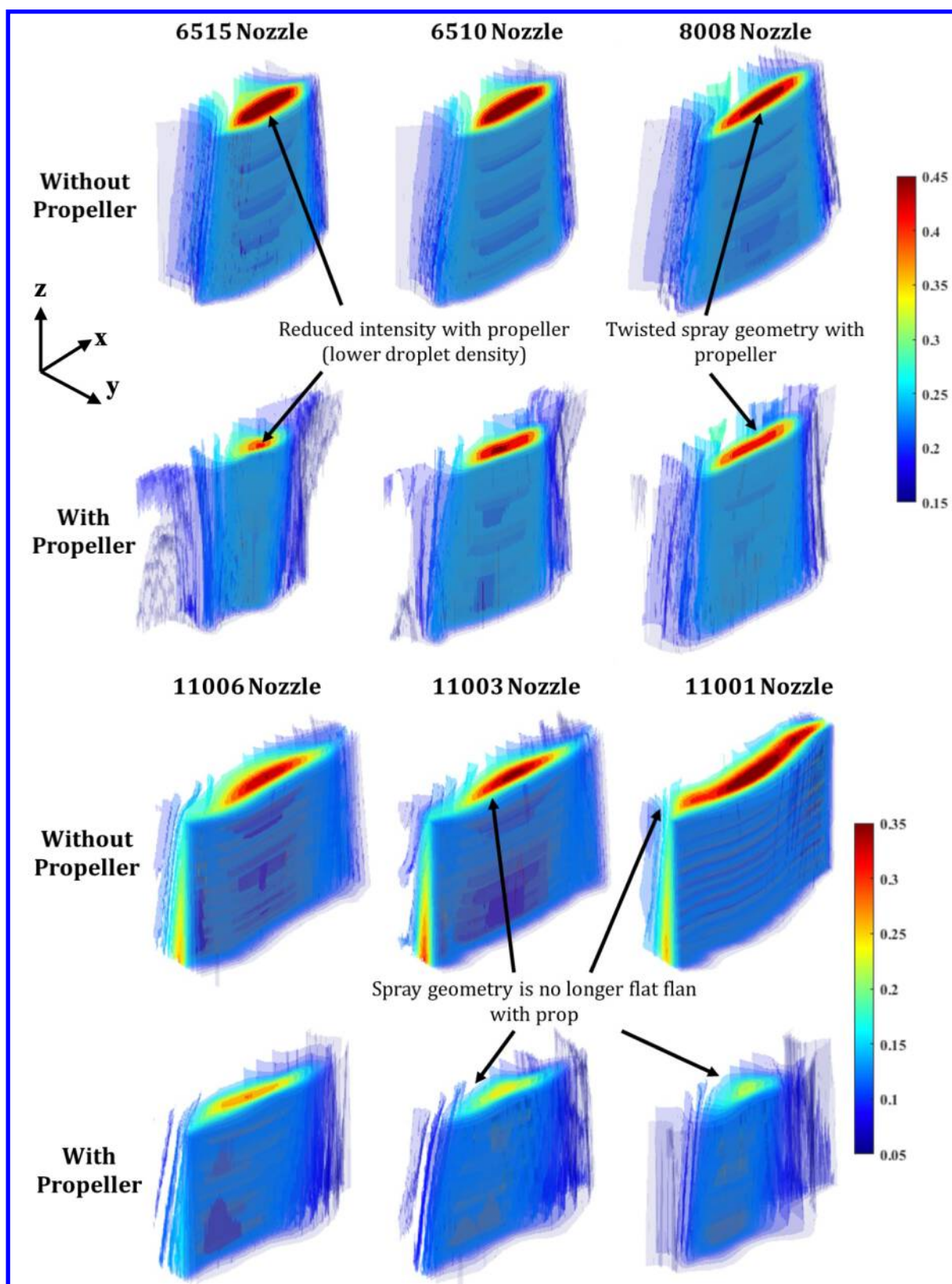


Figure 8 3D reconstructions of spray geometry for coarser nozzles (6515, 6510, and 8008) and finer nozzles (11006, 11003, and 11001) with and without the influence of the propeller. The influence of the propeller introduces significant changes to the spray, including broadening, twisting, and radial ejection of finer droplets.

In the absence of propeller influence, the spray geometry for all nozzles exhibited a concentrated high-density region at the center, with droplet density decreasing towards the spray edges. However, the introduction of the propeller drastically altered the spray geometry. For the 6515 nozzle, the flat fan spray transformed into a more circular geometry. The 6510 nozzle exhibited similar behavior, although the changes were less pronounced compared to the 6515 nozzle. A noticeable twist along the length of the spray was observed for both nozzles, caused by the swirling nature of the propeller wake beneath the hub, as shown in Figure 2. Significant changes were also observed at the outer edges of the spray, where droplet densities decreased and smaller droplets were distributed over a wider area, increasing the potential for spray drift. Among the coarser nozzles, the 8008 nozzle showed the smallest deviations in spray geometry when influenced by the propeller. Although a reduction in droplet density and widening of the spray were still observed for the 8008 nozzle, the extent of these effects was less severe compared to the 6515 and 6510 nozzles.

Similar trends are observed in the finer nozzles shown in the bottom two rows of Figure 8. Due to the wider fan angle of 110° , the outer edges of the spray from these finer nozzles could not be fully captured by the patternator at a distance of 38 inches from the nozzle. At this distance, the spray width exceeds 6 feet, and expanding the laser sheet further would have reduced its overall intensity, making it insufficient for capturing the edges. Without the influence of the propeller, the finer nozzles exhibit a flat fan spray geometry, similar to the coarser nozzles. The highest droplet concentration is observed in the central portion of the spray, with density decreasing toward the perimeter. However, the overall intensity for finer nozzles is lower than that of coarser nozzles, which is attributed to their significantly smaller droplet sizes, as indicated in Table 1.

When influenced by the propeller, the finer nozzles show a much more pronounced change in spray geometry compared to the coarser nozzles. This heightened sensitivity is due to the smaller droplet sizes, which are more easily affected by the turbulent and three-dimensional currents induced by the propeller wake. For the 11003 and 11001 nozzles, the propeller alters the spray geometry so significantly that it no longer resembles a flat fan pattern. Instead, the spray takes on a circular geometry that mirrors the propeller wake, particularly for the 11001 nozzle. The spray boundary spreads in all radial directions, contributing substantially to spray drift. Similar to the coarser nozzles, the finer nozzles also experience a pronounced twist in the spray geometry, forming an S-shaped pattern, as previously observed in the flow visualization in Figure 3. The direction of the twist aligns with the rotational direction of the propeller, further confirming the strong influence of the propeller wake on the finer nozzle sprays.

A traditional method for quantifying droplet sizes in a spray involves the use of laser diffraction instruments, such as Malvern's SprayTech, as shown in Figure 4. This instrument measures the droplet size distribution at 16 inches from the nozzle tip as the spray traverses through the laser's line of sight. The droplet size measurements obtained with and without the influence of the propeller were previously discussed in detail by Tierney et al. [15]. These measurements indicated a shift in the droplet size distribution toward larger droplet sizes for nearly all nozzles. However, gutter tests, shadowgraphy, and Diffused Background Imaging (DBI) from Tierney et al. [15] suggest that this shift is not due to an actual increase in droplet size but rather the exclusion of finer droplets from the measurement. Under the influence of the propeller, finer droplets are ejected radially from the main spray body by the strong three-dimensional currents induced by the propeller wake, causing them to evade the laser diffraction instrument's line of sight. The results in Figure 8 provide further evidence for this claim, showing that the widening of the spray begins well before the measurement plane at 16 inches from the nozzle tip. The topmost section of the spray shown in Figure 8, captured at 10 inches from the nozzle, already reveals finer droplets being ejected radially from the main spray body. This is evident from the unwrapping of the iso-contours at the spray perimeter across all nozzle cases. The radial ejection of finer droplets affects the entirety of the spray geometry, not just the bottom portion. Since the laser diffraction measurements are taken at 16 inches from the nozzle tip, any droplets ejected radially between the nozzle tip and the measurement plane are excluded from the instrument's readings, leading to an apparent but misleading shift toward larger droplet sizes.

The changes in the overall spray width due to the influence of the propeller were quantified by analyzing a slice of the intensity volume from the patternator results along the Y-axis. The resultant intensity distribution across the X-Z plane at $Y=0$ (center of the spray) is shown in Figure 9. Similar to Figure 8, the first two rows represent the coarser nozzles with and without the influence of the propeller, while the last two rows depict the intensity contours for the

finer nozzles under the same conditions. In the contour plots, the red regions indicate areas of higher droplet intensity, whereas the dark blue regions represent negligible intensity, where no droplets were captured by the patterator.

The trends in Figure 9 mirror those observed in Figure 8, with the coarser nozzles, particularly the 6515 nozzle, exhibiting significant changes in spray geometry under the influence of the propeller. The 6515 nozzle demonstrates a marked increase in spray width and a redistribution of droplets, while the 8008 nozzle shows minimal deviation, maintaining a relatively stable width even with the propeller influence. For all nozzles without the influence of the propeller, the intensity contours show a symmetric distribution of droplet densities about the Z-axis. However, this symmetry breaks down under the influence of the propeller, as the twisting motion introduced by the propeller wake creates a pronounced asymmetry. This effect is particularly evident in planes farther from the nozzle tip, where the twist causes a redistribution of droplets and a significant decrease in droplet density along the length of the spray (Z-axis). These results further highlight the disruptive influence of the propeller on both the geometry and density distribution of the spray.

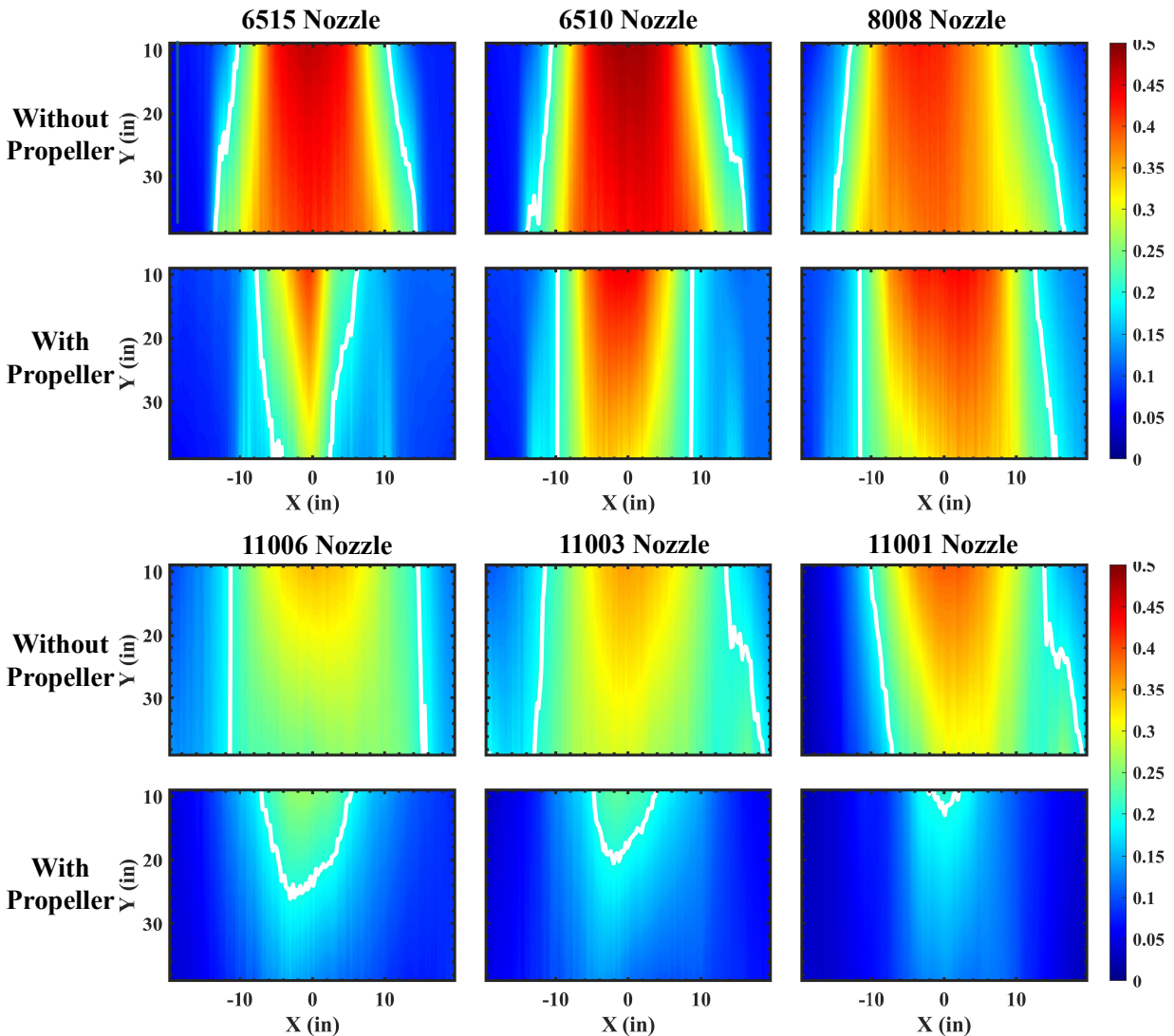


Figure 9 Intensity contours of spray distribution in the X-Z plane at Y=0 for coarser nozzles (6515, 6510, and 8008) and finer nozzles (11006, 11003, and 11001) with and without the influence of the propeller. The contours highlight symmetry without the propeller and asymmetry induced by the propeller wake, particularly for the coarser 6515 nozzle.

Figure 9 also presents the intensity contours for the finer nozzles, with and without the influence of the propeller. Unlike in Figure 8, the intensity range of the contours was kept consistent between the coarse and fine nozzles for comparative purposes. As expected, the intensity of backscattered light from the droplets decreases significantly for the finer nozzles due to their smaller droplet diameters—an order of magnitude smaller than those of the coarser nozzles—and the lower overall droplet density. Notably, the 11001 nozzle exhibits a comparatively higher intensity without the influence of the propeller when compared to the other finer nozzles. However, with the influence of the propeller, the intensity diminishes significantly, accompanied by a pronounced asymmetry about the Z-axis. This comparison highlights that finer nozzles are more heavily affected by the propeller wake than the coarser nozzles, as the smaller droplets are more susceptible to the turbulent, three-dimensional flow induced by the propeller.

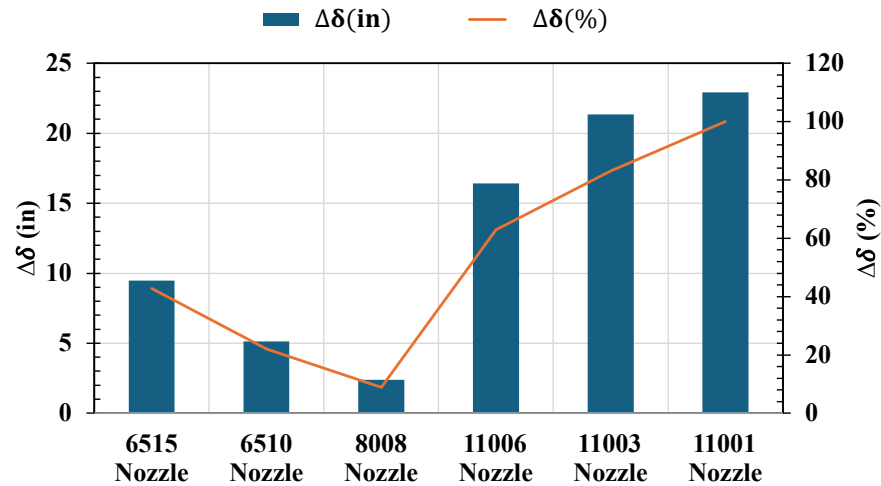


Figure 10 Quantification of spray width changes for all nozzles due to propeller influence, showing absolute changes in inches (blue bars) and percentage changes (orange line).

To quantify the changes in spray width across all nozzles under the influence of the propeller, the spray boundary was defined as the region where the spray intensity reaches a magnitude of 0.2, as indicated by the white contour lines in Figure 9. The distance between the 0.2 intensity regions on both ends of the spray was measured at a Z-location of 16 inches, corresponding to the laser diffraction measurement plane. Figure 10 illustrates the changes in spray width, both in inches and as a percentage, due to the influence of the propeller for all six nozzles.

As shown in Figures 8 and 9, the propeller induces substantial changes in spray geometry, with percentage changes ranging from 10% to 100%. For the coarser nozzles, the percentage change in spray width remains within 10%, with the smallest change observed for the 8008 nozzle and the largest for the 6515 nozzle. In contrast, the finer nozzles exhibit a much greater sensitivity to the propeller's influence, with an average change in spray width of approximately 90%. The 11001 nozzle experiences the most dramatic change, with nearly a 100% increase in spray width, as the finer droplets are significantly dispersed by the propeller's wake, ejecting droplets radially in all directions.

Similar to the changes in spray width, the changes in spray depth were measured by analyzing a slice of the spray volume at the $X=0$ plane. The resultant intensity contours for all six nozzles, with and without the influence of the propeller, are shown in Figure 11. As in previous contour plots, the red regions represent areas of higher droplet density, while the blue regions indicate regions of lower density. For the coarser nozzles, the droplet intensity remains relatively uniform along the length of the spray in the absence of the propeller. The spray depth, defined as the distance between the front and rear ends of the spray where the intensity drops to a magnitude of 0.2, is largest for the 6515 nozzle and decreases for the 6510 and 8008 nozzles. Without the influence of the propeller, the spray depth remains consistent along its length. However, under the influence of the propeller, the spray depth decreases progressively along the spray length due to the swirling flow induced by the propeller, which ejects droplets radially outward and twists the spray pattern, as seen in Figure 8. The intensity magnitude also decreases under propeller influence, indicating a reduction in droplet density.

A slight angular deviation in the spray pattern is also observed under the influence of the propeller, particularly in the finer nozzles. For the finer nozzles, the intensity and depth of the spray are significantly lower compared to the coarser nozzles, as expected. However, the reduction in intensity is more pronounced under the influence of the propeller, demonstrating the finer droplets' increased sensitivity to the turbulent wake.

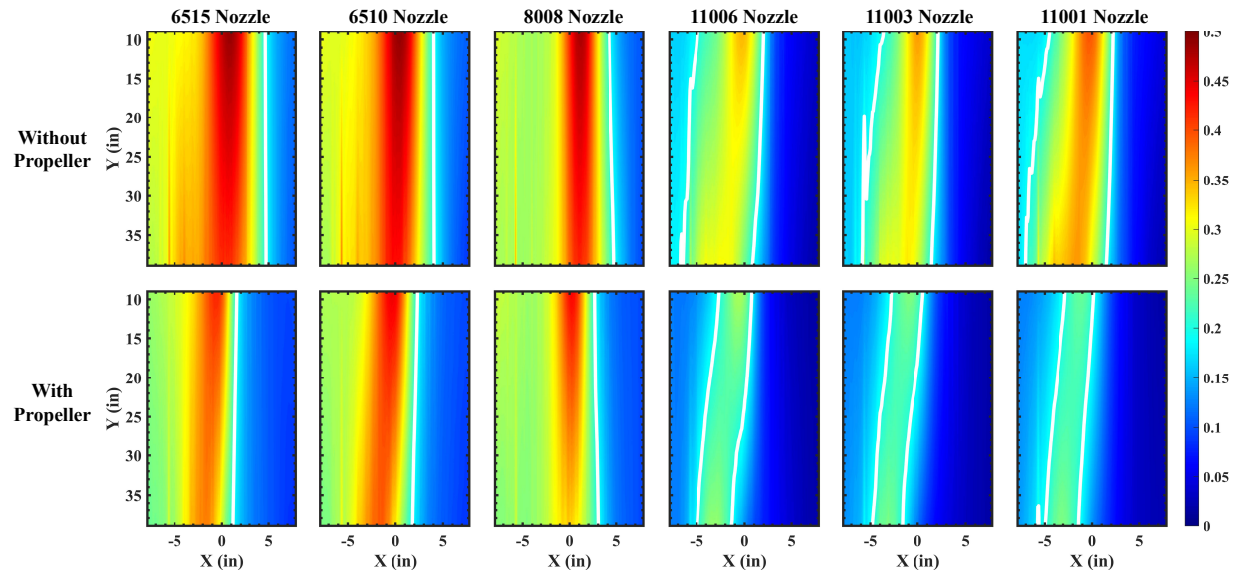


Figure 11 Intensity contours showing the depth of the spray at the $X=0$ plane for coarser nozzles (6515, 6510, and 8008) and finer nozzles (11006, 11003, and 11001), with and without propeller influence. The propeller reduces spray depth, decreases droplet density, and introduces angular deviations, particularly for finer nozzles.

To quantify the changes in spray depth, the distance between the front and rear ends of the spray, where the intensity magnitude reaches 0.2, was measured and is plotted in Figure 12. These depth measurements were taken at 16 inches from the nozzle to align with the laser diffraction measurement plane.

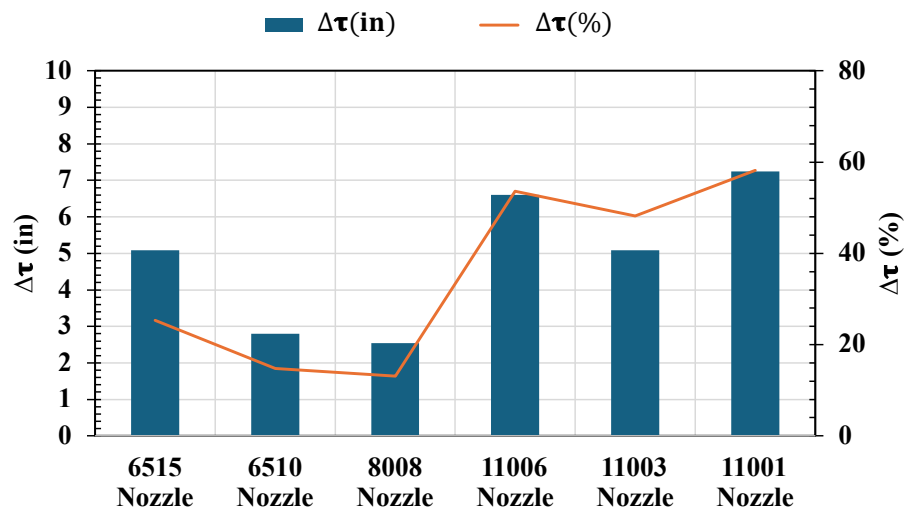


Figure 12 Quantification of spray depth changes for all nozzles at 16 inches from the nozzle tip, with and without propeller influence. The left Y-axis shows the absolute depth change (blue bars) in inches, and the right Y-axis represents percentage change (orange line).

As in Figure 10, the left Y-axis represents the change in spray depth in inches, while the right Y-axis shows the change as a percentage. The trends in spray depth closely resemble those observed for spray width. The finer nozzles exhibit an average change in depth of up to 50%, whereas the coarser nozzles show changes of up to 30%, with the

8008 nozzle exhibiting the smallest change overall. Notably, the finer nozzles exhibit a greater change in width compared to depth under the influence of the propeller, reflecting their increased sensitivity to the swirling and turbulent flow generated by the propeller wake.

To illustrate changes in intensity and geometry across the spray cross-section, a Z-slice was obtained from the spray volume at 16 inches from the nozzle, corresponding to the laser diffraction measurement plane, as shown in Figure 13. The first two rows depict the spray cross-sections for the coarser nozzles with and without the influence of the propeller, while the bottom two rows show the corresponding cross-sections for the finer nozzles. For the coarser nozzles, the flat fan shape of the spray geometry is evident. The left and right edges of the spray align well with the expected 65° spray angle for the 6515 and 6510 nozzles and the 80° spray angle for the 8008 nozzle. Droplet density is highest at the center of the spray and decreases toward the edges for all cases. Under the influence of the propeller, the spray geometry changes significantly. The 6515 nozzle exhibits the most dramatic transformation, with the flat fan geometry shifting to a more circular shape. While the 6510 and 8008 nozzles retain the flat fan nature of their geometry, an S-shaped twist induced by the propeller is clearly observed. Additionally, the overall intensity at the center of the spray decreases under the propeller's influence.

Similar trends are observed for the finer nozzles. Without the influence of the propeller, the spray exhibits a nearly flat fan geometry. However, the introduction of the propeller causes drastic changes in the spray shape. The S-shaped twist becomes readily visible, and droplet intensity decreases significantly. The 11001 nozzle shows the most substantial change, with the flat fan geometry completely lost under the influence of the propeller, transitioning instead to a highly distorted shape.

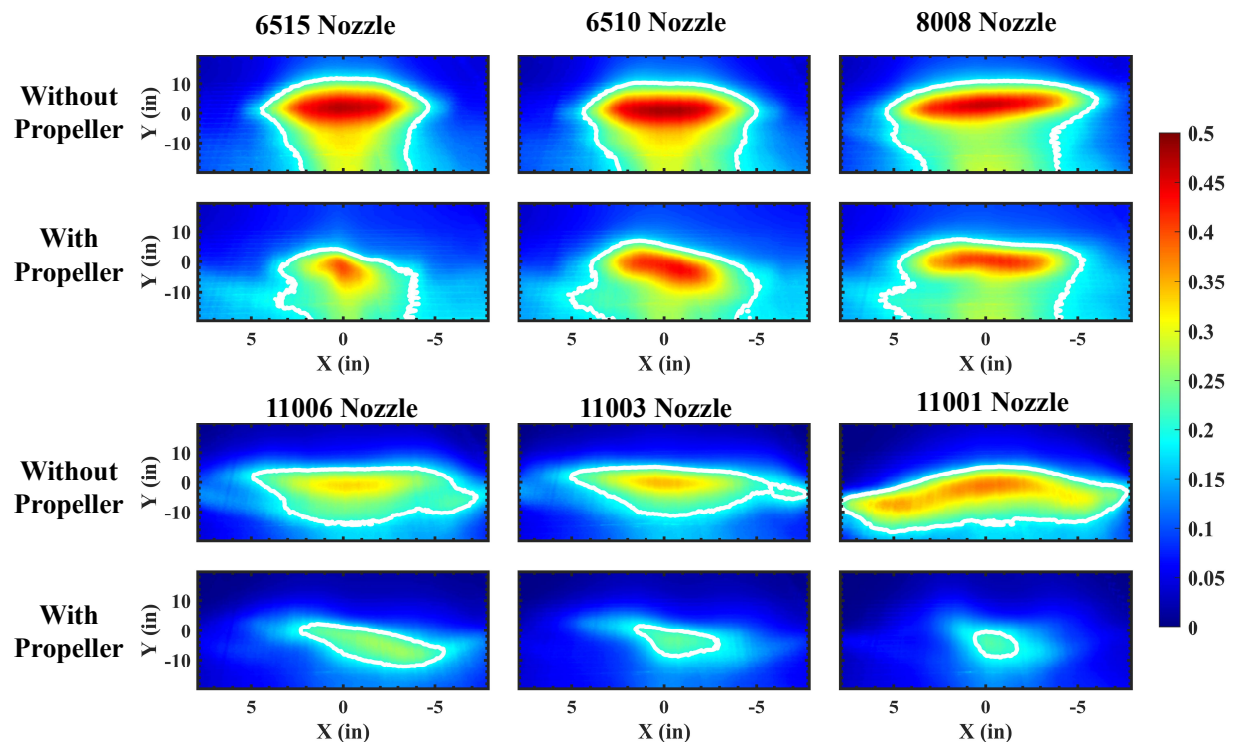


Figure 13 Cross-sectional intensity contours of the spray at $Z=16$ inches for coarser nozzles (6515, 6510, and 8008) and finer nozzles (11006, 11003, and 11001), with and without the influence of the propeller. Coarser nozzles retain more of their flat fan geometry, while finer nozzles, especially the 11001 nozzle, exhibit significant deformation and intensity reduction.

The results from Figure 13 were used to quantify the changes in maximum intensity for all nozzles with and without the influence of the propeller. The change in intensity directly correlates to droplet sizes and droplet density. While the patternator cannot differentiate whether the change in intensity is due to variations in droplet density or droplet

size, it does provide evidence that the spray's nature—including droplet size and density—is significantly altered under the influence of the propeller. Figure 14 illustrates the change in maximum intensity for all nozzles. The left Y-axis represents the change in intensity magnitude, while the right Y-axis shows the change as a percentage. Consistent with the trends observed in Figures 10 and 12, the coarser nozzles exhibit comparatively smaller changes in intensity, averaging around 15%. In contrast, the finer nozzles demonstrate significantly larger changes, with the 11001 nozzle experiencing the most dramatic decrease in intensity. This finding underscores the greater sensitivity of finer nozzles to the propeller's influence, likely due to the smaller droplet sizes being more susceptible to the turbulent wake and radial dispersion induced by the propeller.

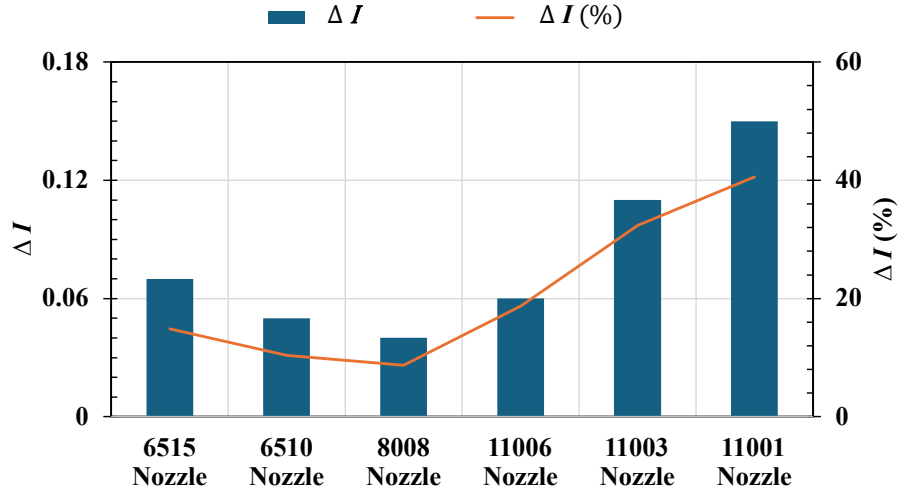


Figure 14 Quantification of changes in maximum spray intensity for all nozzles with and without propeller influence. The left Y-axis represents the change in intensity magnitude (ΔI), while the right Y-axis shows the percentage change. Finer nozzles exhibit greater reductions in intensity, with the 11001 nozzle showing the largest change due to the influence of the propeller.

The results presented indicate that finer nozzles experience significant deviations from their expected spray geometry under the influence of the propeller. To explain this phenomenon, the drag coefficient of the droplets was calculated based on their diameter, exit velocity from the nozzle, and the velocity imparted by the propeller wake. The theoretical model from Holterman [19], shown in Equation 2, was used to estimate C_d :

$$C_d = \left(\left(\frac{a}{Re} \right)^c + b^c \right)^{1/c} \quad (2)$$

where $a = 24$, $b = 0.32$, and $c = 0.52$. This model has a relative error of $\pm 5\%$ when compared to experimental results, as reported in [19]. The Reynolds number (Re) of the droplets was calculated using Equation 3.

$$Re = \frac{\rho_{air} D_{Droplet} V_{Droplet}}{\mu_{air}} \quad (3)$$

Here, the droplet velocity ($V_{Droplet}$) was determined as a combination of the nozzle exit velocity ($V_{Nozzle Exit}$) and the propeller-induced velocity ($V_{Propeller Exit}$), as shown in Equation 4.

$$V_{Droplet} = V_{Nozzle Exit} + V_{Propeller Exit} \quad (4)$$

The nozzle exit velocity ($V_{Nozzle\ Exit}$) was calculated using the volumetric flow rate and the orifice area, derived from the nozzle orifice diameter ($D_{Orifice}$) given in [20]. The orifice area ($A_{Orifice}$) was determined using Equation 5.

$$A_{Orifice} = \pi D_{Orifice}^2 / 4 \quad (5)$$

The volumetric flow rate (Q), converted from GPM (Table 1), and the discharge coefficient (D), assumed to be 0.9 for standard flat fan nozzles [21], were used to calculate $V_{Nozzle\ Exit}$ using Equation 6

$$V_{Nozzle\ Exit} = D * Q / A_{Orifice} \quad (6)$$

The calculated values for $D_{Orifice}$, $V_{Nozzle\ Exit}$, $V_{Droplet}$ and Re for all nozzles are presented in Table 4. Notably, while the 11001 and 11003 nozzles have a similar $D_{Orifice}$, the higher inlet pressure and lower flow rate of the 11001 nozzle result in a lower exit velocity compared to the 11003 nozzle. Similarly, the 6515 nozzle exhibits a significantly lower exit velocity than the 6510 and 8008 nozzles. The Reynolds numbers of the droplets range from 74 to 2137 across the nozzles.

Table 4 Calculated droplet characteristics for various nozzles including $D_{Orifice}$, $V_{Nozzle\ Exit}$, $V_{Droplet}$ and Re . Higher drag coefficients for finer nozzles correspond to lower Reynolds numbers.

Nozzle	$D_{Orifice}$ (m)	$V_{Nozzle\ Exit}$ (m/s)	$V_{Droplet}$ (m/s)	Reynolds Number		
				Dv(10)	Dv(50)	Dv(90)
11001	0.0007	13	20	74	160	275
11003	0.0011	32	37	273	609	1018
11006	0.0015	32	37	400	876	1437
8008	0.0018	37	41	549	1208	1756
6510	0.0020	32	37	592	1299	2137
6515	0.0028	12	19	402	859	1424

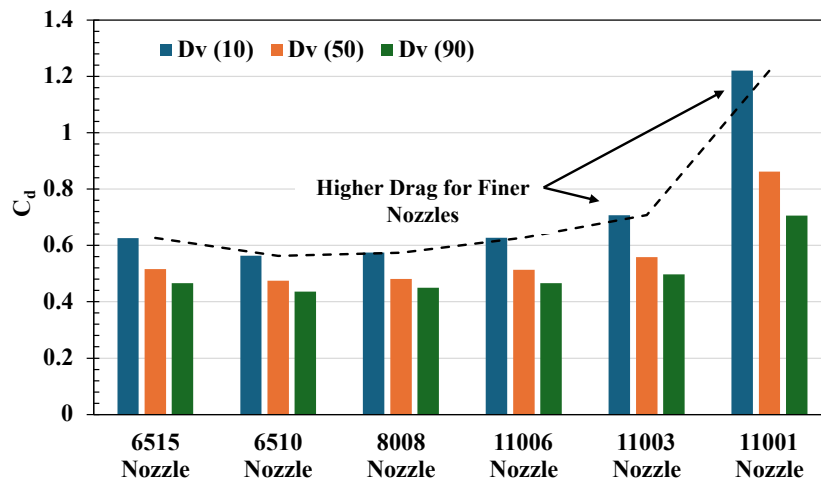


Figure 15 Drag coefficients (C_d) for droplets produced by various nozzles, calculated using Holterman's model [19]. Lower Reynolds numbers for finer nozzles result in higher drag coefficients, explaining their greater sensitivity to propeller-induced disturbances.

The resultant drag coefficients (C_d) for the droplets, calculated using Equation 2, are shown in Figure 15. The lower Reynolds numbers of droplets from finer nozzles result in higher drag coefficients, whereas the larger Reynolds

numbers of coarser nozzle droplets correspond to lower drag coefficients. The higher drag coefficients associated with finer nozzles explain their greater deviations in spray geometry under the influence of the propeller. Additionally, the 6515 nozzle exhibits a slightly higher drag coefficient compared to the 6510 and 8008 nozzles, which may account for its larger observed deviations in spray geometry, as seen in Figure 8. Overall, the trends in drag coefficient variation align with those observed in spray geometry, further validating the use of patternator visualization to assess spray behavior.

IV. Conclusion

This study investigated the impact of propeller-induced wake on the spray geometry and droplet behavior of agricultural nozzles using a combination of patternator-based visualization and theoretical drag coefficient calculations. The results demonstrate significant alterations in spray characteristics under the influence of the propeller, with finer nozzles showing greater sensitivity compared to coarser nozzles.

The patternator results revealed that the propeller wake induces considerable changes in spray width, depth, and cross-sectional geometry. Finer nozzles, such as the 11001 and 11003, exhibited up to a 100% increase in spray width and 50% reduction in depth due to the strong radial dispersion and S-shaped twisting caused by the turbulent, three-dimensional flow of the propeller wake. In contrast, coarser nozzles like the 8008 exhibited minimal deviations, retaining much of their flat fan geometry despite the propeller's influence.

Quantification of droplet dynamics further supports these findings. The finer nozzles, with smaller droplet sizes and lower Reynolds numbers, were found to have higher drag coefficients, making them more susceptible to disturbances from the propeller wake. Coarser nozzles, with larger droplet sizes and higher Reynolds numbers, experienced lower drag coefficients, resulting in greater stability under propeller influence. This correlation between drag coefficients and spray geometry deviations underscores the physical basis for the observed trends.

Overall, this study highlights the need to account for propeller wake effects in the design and operation of agricultural UAV spraying systems, particularly when selecting nozzles for specific applications. The findings suggest that finer nozzles, while capable of producing smaller droplets for precise spraying, are more prone to drift and distortion in the presence of propeller wake, raising concerns about spray efficiency and environmental safety. Future work should explore methods to mitigate these effects, such as nozzle positioning, shield designs, or improved flow control strategies.

V. References

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